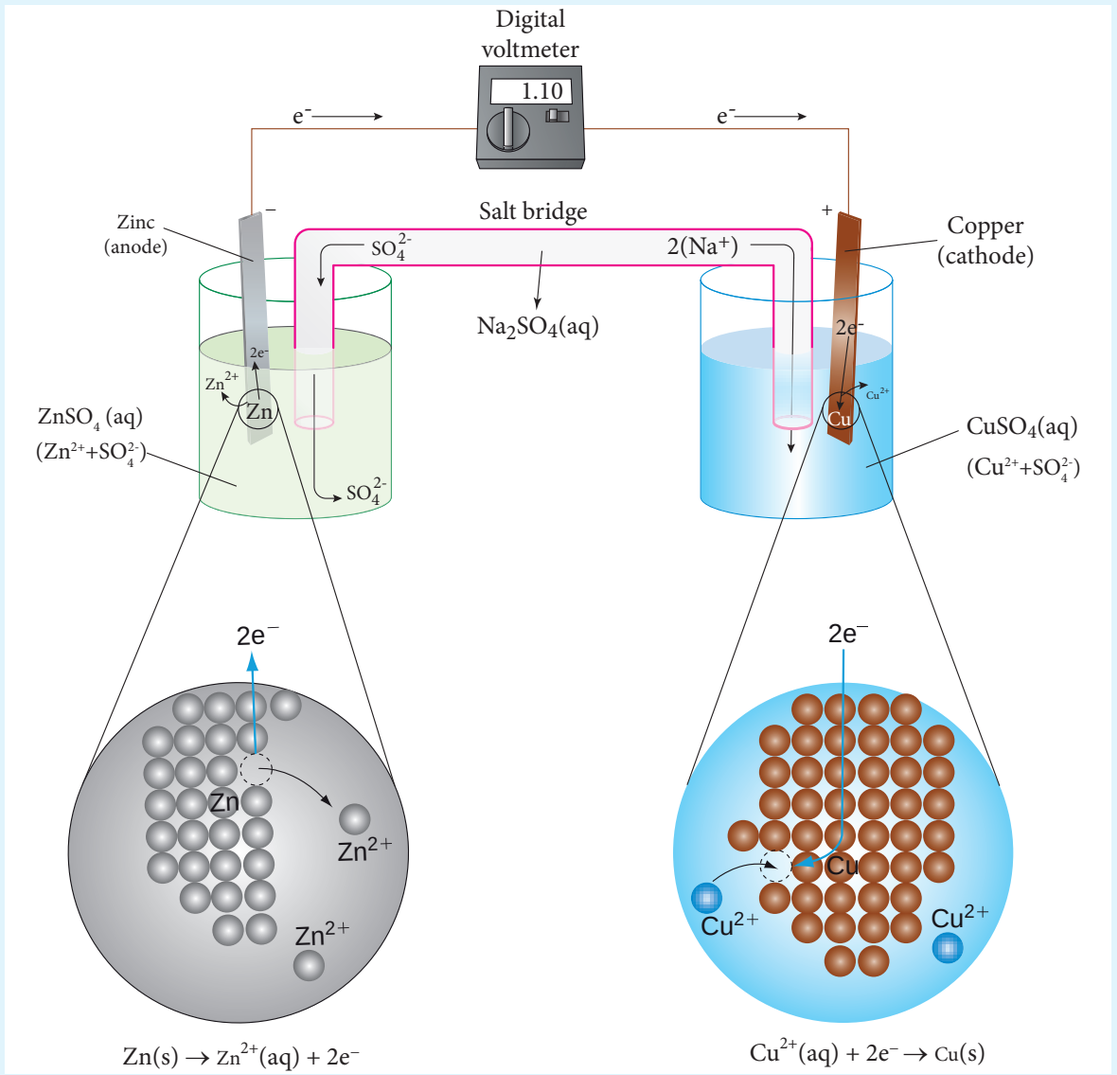


Figure 9.5 : Daniel cell

Anodic oxidation

The electrode at which the oxidation occurs is called the anode. In Daniel cell, the oxidation take place at zinc electrode, i.e., zinc is oxidised to Zn^{2+} ions by loosing its electrons. The Zn^{2+} ions enter the solution and the electrons enter the zinc metal, then flow through the external wire and then enter the copper strip. Electrons are liberated at zinc electrode and hence it is negative (- ve).

$$Zn(s) \rightarrow Zn^{2+}(aq) + 2e^{-} \qquad \text{(loss of electron-oxidation)}$$

Cathodic reduction

As discussed earlier, the electrons flow through the circuit from zinc to copper, where the Cu^{2+} ions in the solution accept the electrons, get reduced to copper and the same get deposited on the electrode. Here, the electrons are consumed and hence it is positive (+ve).

$$Cu^{2+}(aq) + 2e^{-} \rightarrow Cu\ (s) \qquad \text{(gain of electron-reduction)}$$

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